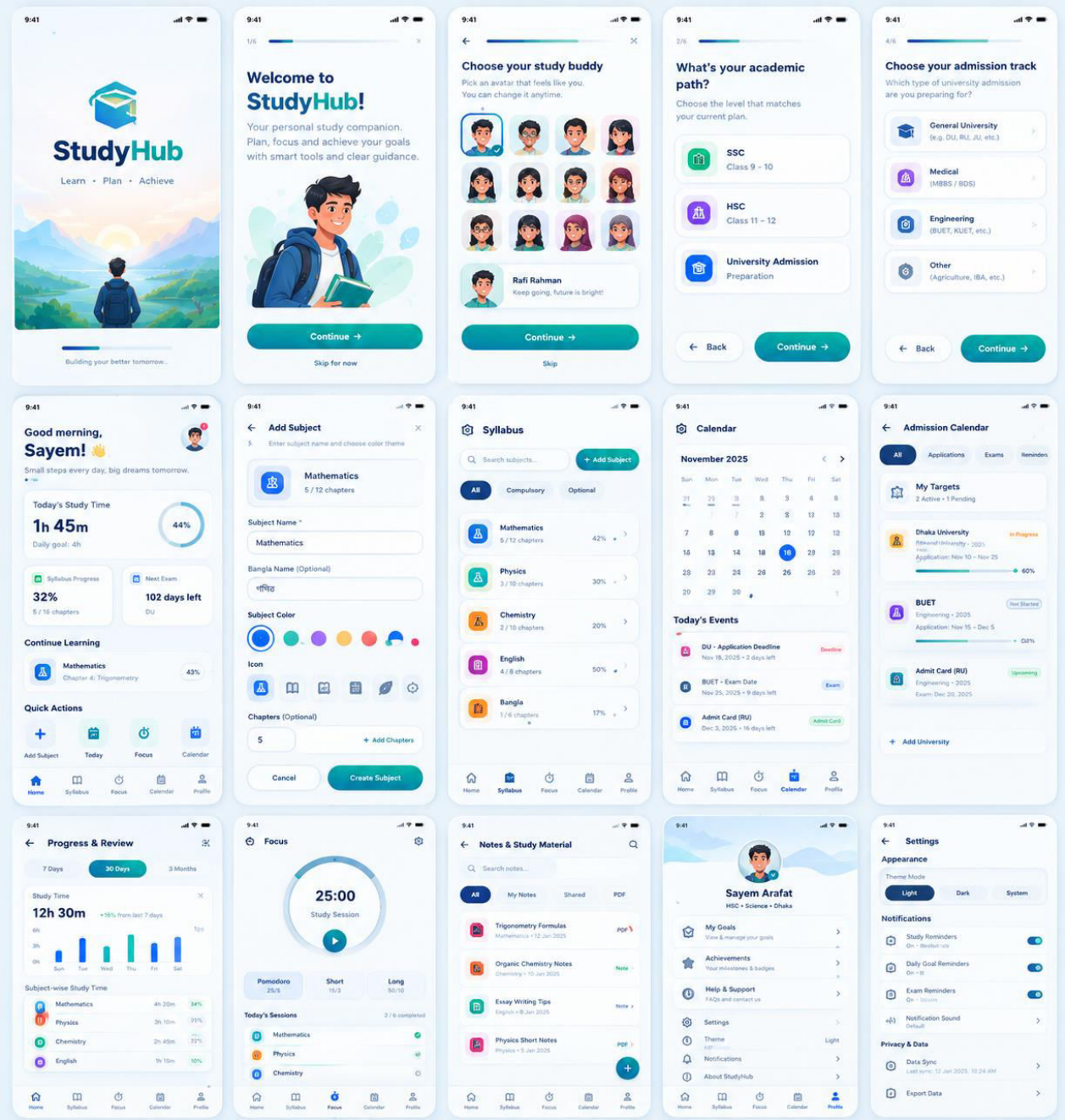
STUDYHUB

Master Product Requirements & Premium UI Specification

Android-first study planner for SSC, HSC and university admission candidates in Bangladesh



VISUAL SOURCE OF TRUTH • ALL SCREENS MUST REMAIN LIGHT MODE

Version 2.0 | Product + UI/UX + Functional Requirements + Acceptance Tests

Document control

Field	Specification
Product	StudyHub
Document type	Master PRD + design system + functional implementation specification
Target platform	Android-first mobile app; adapt to the actual existing project's framework
Primary users	Bangladesh SSC students, HSC students, and university admission candidates
Visual reference	The embedded all-light-mode StudyHub UI concept image in this document
Priority	High-fidelity visual consistency and reliable real functionality
Version	2.0 — premium UI and functional expansion
Source-of-truth order	This PRD defines behavior; the embedded mockup defines visual direction; existing working code/data must be preserved

How to use this document

- Give this entire document and the embedded reference image to the coding agent as the primary implementation brief.
- The mockup is a visual target, not a static screenshot to paste into the app. Rebuild each screen with real, responsive UI components.
- Treat all screens as one product: same typography, color system, spacing, card language, icon family, top-bar rules, and bottom navigation.
- Do not create dark-mode screens in the redesign. The required visual deliverable and every screen in this reference must be LIGHT MODE. If the app already has a dark-theme feature, it may remain a separate user-selectable capability, but do not mix dark surfaces into the light UI or the reference montage.
- Never replace functional behavior with fake demo content. Empty states are preferred over invented data.

1. Executive summary

StudyHub is a premium, dependable study-management companion for students in Bangladesh. It helps learners organize academic information, subjects, chapters, daily goals, study sessions, exams, admission deadlines, notes, reminders, and progress in one coherent app.

The redesign must deliver the visual quality of a polished, high-end mobile product: refined layout, clean typography, a confident StudyHub identity, carefully balanced white surfaces, restrained teal/blue accents, soft shadows, precise alignment, friendly custom cartoon avatars, and smooth but restrained motion. Visual polish is not enough: every button, form, date, progress indicator, notification, and navigation item must be connected to real application state.

Product promise

Know what to study, know what is coming next, and see honest progress—without clutter, fake metrics, or repeated setup screens.

Non-negotiable outcomes

- All redesigned reference screens use one consistent LIGHT visual system.
- No hamburger menu in the primary app top bar.
- A premium, bold StudyHub wordmark and a distinctive vector-friendly brand mark.
- A real first-launch onboarding flow that is not repeated on every app launch.
- A stable library of cohesive cartoon avatars; selected avatar persists across app restarts.
- Real subject creation, chapter tracking, study planning, focus sessions, analytics, and admission-calendar records.

- Notifications use an honest permission and scheduling flow; no claim of notification delivery without platform support.
- No fabricated student names, progress percentages, streaks, exam dates, university deadlines, or study records in production.
- Correct Android status-bar, keyboard, gesture, safe-area, and bottom-navigation behavior.
- The app builds and critical user journeys are tested before the work is reported as complete.

2. Product goals, non-goals, and success measures

Goals

- Help students establish a personalized academic profile with minimal friction.
- Turn study intentions into scheduled tasks and measurable study sessions.
- Show syllabus completion based on chapters the student actually tracks.
- Help admission candidates track universities, application windows, deadlines, admit-card dates, and exam dates.
- Offer fast access to focus timer, notes, calendar, and progress review.
- Make the app feel premium and coherent on common Android phones.
- Keep essential personal study data available offline where the current architecture supports local persistence.

Non-goals for the initial release

- Do not build a social network, public leaderboard, chat platform, or complex classroom LMS unless explicitly added to a later roadmap.
- Do not invent official university admission dates or claim that sample dates are verified.
- Do not require a backend merely to render the core UI if local-first functionality can meet the requirement.
- Do not use a screenshot as the app screen, or substitute static charts for real calculations.
- Do not perform a framework migration unless the existing project is unworkable and the reason is clearly documented.

Measure	Expected behavior / acceptance
Onboarding completion	Profile setup saves successfully; returning users bypass first-run onboarding.
Navigation	All primary destinations open without dead ends or unexpected resets.
Data integrity	Created subjects, chapters, tasks, study sessions, and admission entries survive app restart.
UI consistency	All screens use shared tokens and components; no accidental one-off colors or spacing.
Theme consistency	The delivered reference and redesigned screens remain light mode; no dark screen appears in the light UI flow.
Notification honesty	Permission denial is handled gracefully; app does not claim reminders are enabled when scheduling is unavailable.
Performance	Scrolling, selection, bottom sheets, and screen transitions feel responsive without visible layout jumps.
Quality	No known critical crashes, blocking errors, dead primary buttons, or major overflow in tested flows.

3. Users and academic pathways

User type	Needs	Conditional setup
SSC student	Subjects, chapter progress, daily routine, exam countdown, reminders	SSC; class/year; group if applicable; target exam year; daily goal
HSC student	Subject and chapter tracking, study planner, focus sessions, HSC exam countdown	HSC; Science / Humanities / Business Studies; year; daily goal

General university admission	Multiple university exams, application windows, deadlines, unit/faculty, exam preparation	Admission; target universities; units; application and exam milestones
Medical admission	Medical admission timeline, application, admit card, exam, biology/chemistry preparation	Medical; relevant exam cycle; targets; important dates
Engineering admission	Engineering university targets, units, application deadlines, exams	Engineering; target institutions; units; important dates
Other admission candidate	Flexible institution and deadline tracking	Other; custom institutions, programs, dates, and notes

The interface must adapt to the selected academic path. Do not show irrelevant setup fields or hardcode a single student's profile into the app. Users must be able to edit academic details later.

4. Visual source of truth and fidelity rules

Reference image rules

- The embedded StudyHub collage is the highest-priority visual reference for the redesign.
- All showcased screens are LIGHT MODE: soft near-white/pale-blue backgrounds, white surfaces, navy text, and teal/blue accents.
- Do not introduce dark-mode screens, black cards, or mixed light/dark panels into the redesigned flow or any new mockup.
- Match the overall structure, spacing rhythm, card proportions, typography hierarchy, button styling, and bottom navigation as closely as is practical.
- Use the image to understand the visual language; implement actual responsive components instead of tracing screenshot pixels.
- Where the image contains small or imperfect generated text, use correct product copy from this PRD rather than reproducing visual-generation errors.
- Do not introduce unrelated branding, a generic cube logo, random icon styles, or inconsistent illustration styles.

Quality bar

A screen is not finished just because its elements exist. It must be composed with intentional spacing, strong alignment, readable text, realistic density, balanced white space, consistent card edges, and polished states for loading, empty, selected, disabled, error, and success conditions.

Visual review checklist

- Does the screen look like it belongs to the same StudyHub product as the other screens?
- Are the title, supporting copy, content, and actions visually ordered correctly?
- Are left/right margins, card radii, field heights, icon sizes, and gaps consistent?
- Is the most important action obvious without overusing saturated color?
- Are all labels readable at typical Android font scaling?
- Does the screen remain polished when there is no data or long text?
- Are bottom navigation and system insets correct?

5. Design system specification

5.1 Color tokens

Token	Suggested value	Usage
Brand / primary teal	#0796A3	Primary brand accent, selected states, key progress and focus actions

Brand / deep teal	#087F8C	Primary button or brand details where contrast is needed
Brand / blue	#2563EB	Secondary action, links, selected calendar date, charts
Text / primary	#142B4A	Main headings, values, key labels
Text / secondary	#66768A	Supporting copy, metadata, helper text
Background / app	#F4F8FC	Main page background; keep subtle and consistent
Surface / card	#FFFFFF	Cards, sheets, input surfaces
Border / subtle	#E2EAF2	Card outlines, dividers, input outlines
Success	#159A75	Completed chapters, successful states
Warning	#F59E0B	Upcoming deadline or caution
Danger	#E6536B	Overdue or destructive/error state
Info	#3B82F6	Informational status and links
Disabled	#AAB6C4	Disabled controls and inactive states

These are starting tokens, not permission to scatter hex values throughout the code. Centralize them in the existing framework's theme system and adjust only when a contrast or implementation review justifies it.

5.2 Typography

Role	Guidance
Brand wordmark	Bold / extra-bold, custom spacing; “StudyHub” exact spelling
Page title	22–28sp equivalent; bold, short, high contrast
Section title	17–20sp equivalent; semibold/bold
Card title	15–17sp equivalent; semibold
Body	14–16sp equivalent; comfortable line height
Secondary metadata	12–14sp equivalent; never too pale to read
Numeric metrics	Large and tabular where possible; align values consistently
Bangla text	Use a font with reliable Bengali glyph coverage and proper line height

Use one coherent font family where supported, with a clear weight scale. Avoid mixing many font families. Test English and Bangla labels, long university names, and enlarged accessibility text.

5.3 Spacing and geometry

Token	Suggested value	Guidance
Base spacing	4dp	All spacing should derive from a small consistent scale.
Compact gap	8dp	Icon-label gap, dense metadata.
Standard gap	12–16dp	Card internal spacing and component separation.
Section gap	20–24dp	Between major content groups.
Screen horizontal padding	16–20dp	Keep aligned across main screens.
Card radius	18–24dp	Rounded, premium, not excessively pill-shaped.
Button height	48–56dp	Comfortable touch target.
Input height	52–60dp	Consistent, spacious fields.
Icon size	20–24dp	Standard navigation and action icons.

5.4 Surfaces, borders, and shadows

- Prefer white surfaces with a thin subtle border over heavy shadows.
- Use shadows sparingly to establish elevation, especially for floating sheets or key surfaces.
- Use one radius family and avoid arbitrary corner values on each screen.

- Do not apply blur or glass effects everywhere; use only when it adds clarity.
- Keep card content aligned to a common internal padding grid.

5.5 Iconography

- Use one consistent icon family, stroke weight, and visual scale.
- Avoid mixing filled, outlined, emoji, and unrelated third-party icon styles randomly.
- Use a filled or tinted active navigation icon and a restrained inactive icon treatment.
- Every icon-only action must have an accessibility label and a clear hit target.

5.6 Logo and wordmark

- Create a custom StudyHub mark related to learning, focus, progress, or a book/knowledge motif.
- The mark must be vector-based where possible, crisp at small sizes, and consistent in splash, header, profile, and app icon.
- Use a bold, polished StudyHub wordmark; exact spelling and capitalization must not vary.
- No hamburger icon in the main app header.
- Keep notification access in the header with a subtle unread dot/count only when there are actual unread notifications.

5.7 Motion system

Interaction	Guideline
Tap feedback	Immediate subtle pressed state; do not delay action.
Screen transition	Short, consistent transition; avoid excessive zoom or bouncing.
Bottom sheet	Smooth slide/fade with clear drag handle if supported.
Progress update	Animate value changes gently without implying fabricated activity.
Selection	Clear selected border/fill/checkmark and accessible selected state.
Loading	Use restrained progress indicator or skeleton only when work is actually pending.
Reduced motion	Respect OS accessibility preference when supported.

Use motion to clarify cause and effect, not as decoration. Never let animation block navigation, form submission, or the focus timer.

6. Information architecture and navigation

Primary bottom navigation

Destination	Purpose	Primary content
Home	Daily overview	Study time, daily goal, syllabus summary, next exam/admission event, continue learning, quick actions
Syllabus	Subjects and chapter progress	Subject list, chapter checklist, progress by subject and academic level
Focus	Timed study sessions	Pomodoro/custom timer, current session, session history
Calendar	Tasks and academic dates	Daily/weekly/monthly calendar, study tasks, exams, admission deadlines
Profile	Personal setup and preferences	Avatar, academic profile, goals, achievements based on real data, settings, notification preferences

Keep the primary navigation stable and visible on main screens. Secondary areas such as Notes, Admission Calendar, Analytics, and Settings can be opened from the relevant destination or Profile. Do not overload the bottom bar with too many items.

Header rules

- The Android status bar remains at the very top.
- The StudyHub top bar appears immediately below the status bar / safe inset.
- Use the StudyHub logo and wordmark at left, notification entry at right, and no hamburger.
- Avoid a second redundant title row when the branded top bar already provides the page identity; detail screens may use a back arrow and a concise title.
- Header height and horizontal alignment must remain consistent across primary screens.

Navigation behavior

- Switching primary tabs must preserve each tab's reasonable scroll and state where the framework supports it.
- Opening a detail screen must provide a clear back path.
- Onboarding, modal forms, and focused full-screen flows may hide bottom navigation when it improves focus.
- Never place bottom navigation at the top or behind Android's gesture bar.
- Use safe-area-aware padding; content must never be obscured by the bottom navigation.

7. Complete screen inventory

ID	Screen / flow	Required content and behavior
S01	Splash	Brand mark, StudyHub wordmark, quiet transition; no fake loading delay.
S02	Welcome	Concise value proposition, illustration, Continue / Get started.
S03	Avatar selection	Stable cartoon avatar gallery, selected preview, save selection.
S04	Academic level	SSC / HSC / University Admission cards.
S05	Academic details	Relevant group, year, academic context based on selected level.
S06	Admission track	General University / Medical / Engineering / Other, conditional to admission.
S07	Daily goal	Hours/minutes or clear presets; validate reasonable values.
S08	Study schedule	Optional preferred days/time windows and reminder preferences.
S09	Onboarding summary	Review selections, back/edit, finish.
S10	Home	Real metrics, syllabus progress, next event, continue learning, quick actions.
S11	Subject list	Search/filter, create/edit/delete with confirmation, empty state.
S12	Add/Edit Subject sheet	Subject name, optional Bangla name, color, icon, chapter count/chapters, preview.
S13	Subject detail	Chapter list, progress, notes link, subject-specific sessions.
S14	Chapter detail/status	Not started / In progress / Completed, optional notes, update progress.
S15	Syllabus	Academic level, subjects, completed/total counts, search/filter.
S16	Study planner	Task create/edit, date/time, duration, subject/chapter, status.
S17	Calendar	Month/week/day views, event markers, real task and exam data.
S18	Admission calendar	Targets, application windows, deadlines, exam dates, reminders, status.
S19	Admission detail	Institution, unit/program, key dates, application status, notes and next action.
S20	Focus timer	Pomodoro/custom duration, start/pause/resume/finish, session logging.

S21	Study session history	Date, duration, subject, completed/abandoned status.
S22	Notes & study material	Create/edit/search/filter notes, subject links, tags, optional attachments.
S23	Progress & review	Study-time chart, goal adherence, chapter completion, subject breakdown.
S24	Notification center	Real notification history/read state if supported; clear empty state.
S25	Profile	Avatar, name, academic details, goals, accurate summary metrics.
S26	Settings	Theme preference if supported, notifications, reminder schedule, data/export controls.
S27	Permission rationale	Explain notifications before platform permission request.
S28	Empty/error/loading states	Designed states for each relevant feature; never fill gaps with fake content.

8. Detailed functional requirements

8.1 Splash and first-launch routing

- Show the StudyHub mark and wordmark with a restrained visual transition.
- Do not display a repeated Get Started screen on every launch.
- Read the saved onboarding-complete/profile state before routing.
- First launch: route to onboarding. Returning user with valid profile: route to Home or last appropriate primary destination.
- If data loading fails, show a recoverable error or continue with safely available local data; do not loop endlessly on splash.
- Splash animation must be short and must not be used to hide slow startup work.

8.2 Onboarding and profile setup

Step	Inputs	Validation and behavior
Welcome	Intro only	Continue moves to avatar/profile flow; Skip behavior must be intentional and not bypass required setup.
Avatar	Choose from bundled/cached set	Persist stable avatar ID, not a temporary random selection.
Academic level	SSC, HSC, Admission	Required; selection changes subsequent fields.
Academic details	Group, target year, institution/program context as relevant	Only relevant fields are shown; validate year and required values.
Admission track	General University, Medical, Engineering, Other	Only shown for Admission; allow multiple targets where appropriate.
Daily goal	Target hours/minutes per day	Require valid range; allow editing later.
Study schedule	Days/time windows, optional reminder choices	Optional; clearly explain reminders and permission behavior.
Review	Summary of entered information	Allow back/edit; final save must be atomic or recoverable.

- Show step progress such as 2/6 with a subtle progress line.
- Maintain input state when the user moves backward and forward.
- Allow the user to change academic level later; explain if existing subjects/tasks may need review rather than silently deleting them.
- Do not hardcode a person's name or sample academic profile as the actual user's data.

- After completion, mark onboarding complete only after the profile has been saved successfully.

8.3 Cartoon avatar system

- Provide a curated set of consistent original cartoon avatars as local assets or reliably cached resources.
- Each avatar has a stable ID, asset reference, accessibility label, and optional display name.
- Use a clear selection ring/check state and live profile preview.
- Save selected avatar ID in profile storage; restore it on app restart.
- Profile edit must reopen the avatar picker and save changes.
- If an asset fails, use a bundled fallback avatar illustration, not a letter-only placeholder unless no illustration can render.
- Do not download untrusted images or generate random avatars on every launch.

8.4 Subject management

Subject creation must look like the reference's polished add-subject sheet: a rounded top sheet with a drag handle (where supported), strong title and subtitle, live card preview, spacious form fields, color swatches, consistent icon choices, optional Bangla name, chapter setup, and a clear primary Create Subject action.

- Fields: subject name (required), Bangla name (optional), color, icon, optional initial chapters or chapter count.
- Update the preview immediately as the name, color, or icon changes.
- Validate non-empty name, trim whitespace, and prevent accidental duplicate subjects or warn clearly.
- Create, edit, and delete must update the syllabus list and Home summaries.
- Deleting a subject must explain consequences for linked chapters/tasks/sessions and require confirmation.
- Use the selected color as a restrained subject accent, not as a full-screen background.
- Create Subject button must persist real data and close only after success.
- If saving fails, keep entered values and show a useful error.

8.5 Syllabus and chapters

- Show actual subjects and chapter totals, including completed/total count and calculated percentage.
- Chapter states: Not Started, In Progress, Completed.
- User can change status through an accessible control and see updated subject and overall progress.
- Support add/edit/reorder chapters where supported by the data model.
- Filter by subject or status and search where the list becomes large.
- Empty state guides the user to add a subject or chapter.
- Never show 100% progress unless all counted chapters are completed; define zero-chapter behavior as empty/not set up, not 100%.

8.6 Home dashboard

- Greeting uses the saved user's preferred name and selected avatar.
- Today's study time is calculated from completed or active logged study sessions according to a documented rule.
- Daily goal and percentage derive from saved goal and today's total duration; cap visual progress at 100% while preserving the actual time value.
- Syllabus progress derives from actual chapter records.
- Next exam/event displays the nearest valid future date from saved data; if no event exists, prompt the user to add one.
- Continue Learning points to the most recently active subject/chapter or shows a useful add-subject empty state.
- Quick actions: Add Subject, Today/Planner, Start Focus, Add Exam/Event. Every action must navigate to a real flow.
- Cards must be visually balanced; avoid unnecessary oversized cards or duplicate metrics.

8.7 Planner and tasks

- Create a task with title, date, optional start time, duration, subject/chapter link, priority, notes, and status.
- Statuses: Pending, Completed, Cancelled; overdue state is derived from date/time and status.

- Task completion updates planner views and can optionally contribute to completion summaries, but must not fabricate study duration.
- Support edit, duplicate if useful, delete with confirmation, and filter by date/status.
- Prevent invalid duration and date values.
- Use date/time formatting appropriate to Bangladesh while respecting device locale.
- Recurring tasks, if implemented, must create predictable instances and avoid duplicate generation.

8.8 Calendar

- Provide month view and a usable selected-day agenda; week/day view may be included if supported by the existing architecture.
- Show distinct but accessible markers for tasks, exams, application deadlines, admit-card dates, and exam dates.
- Selecting a day updates the agenda below without resetting unrelated state.
- Events must open their real detail screen.
- Support empty days and months without events.
- Calendar navigation must handle month/year boundaries and leap years.

8.9 Admission calendar — essential differentiator

Admission calendar is a first-class feature for university admission candidates. It must let students track multiple institutions and the entire timeline for each target, not just show a single countdown card.

Data field	Requirement
Institution	Name; optional category such as general university, medical, engineering, other.
Program/unit	Unit, faculty, group, or program name; optional when not applicable.
Admission cycle	Academic year/cycle label.
Application opens	Date, optional time, and source note.
Application deadline	Date/time and deadline status.
Admit card available	Optional date/time.
Admission exam	Date/time, location or instructions if user supplied.
Result date	Optional planned/official date.
Application status	Interested, Planning, Application Open, In Progress, Submitted, Admit Card Available, Exam Scheduled, Exam Completed, Result Published, Not Applying.
Progress checklist	User-defined steps such as account created, fee paid, form submitted, documents prepared.
Notes and links	User-entered official URL, notes, eligibility reminders.
Reminder preferences	Per-milestone reminder settings, subject to notification permission and platform support.

Admission calendar behavior

- Allow multiple university/institution targets at once.
- Add a target through a polished bottom sheet/form similar in quality to the reference add-subject UI.
- Show upcoming events ordered by date and a calendar with distinct event markers.
- Show application status, days remaining, progress checklist, and next action for each target.
- Calculate days remaining from actual saved dates using the device's local date; do not hardcode a countdown.
- Allow a candidate to edit, archive, or remove a target with confirmation.
- Mark date confidence/source: user-entered, manually verified, or imported from a trusted source when that capability exists.
- Never claim official dates are current or verified unless they were actually verified and the source is stored.
- If no admission target exists, show an elegant empty state and Add University action.
- If the user selected SSC/HSC only, do not force admission calendar setup; keep it accessible for later.

8.10 Focus timer and study sessions

- Support default Pomodoro 25-minute focus / 5-minute short break and user-configurable durations.
- Controls: Start, Pause, Resume, Finish, Cancel. State transitions must be deterministic.
- Display remaining time clearly, with an animated ring that matches the light theme.
- Optionally associate a session with a subject/chapter/task.
- On session completion, ask for confirmation only where needed and save actual elapsed focus time.
- Interrupted or cancelled sessions must not be recorded as fully completed without an explicit policy.
- Prevent duplicate session records when the user taps Finish repeatedly.
- Persist enough timer state to handle app backgrounding/reopening safely, within platform limits.
- Use notifications or foreground service only where appropriate and supported; request permissions and explain behavior honestly.

8.11 Notes and study materials

- Create, edit, search, and delete notes.
- Optional subject association, tags, favorite/pinned state, and attachment metadata if supported.
- Support autosave or explicit Save with visible save status.
- Handle long content, keyboard insets, and accidental navigation away from unsaved edits.
- Empty state must invite the user to create the first note.
- Do not show fake PDFs or sample notes as if the user created them.

8.12 Progress & review

- Provide selectable periods such as 7 days, 30 days, and 3 months.
- Calculate total study time from saved sessions within the selected period.
- Show daily/weekly goal progress based on stored goals and real session data.
- Show chapter completion counts and subject-wise time distribution.
- Any consistency score or streak must have a documented calculation and must use actual activity.
- Empty charts must have a meaningful zero-data state; do not draw fake bars.
- Chart labels, units, and values must be accessible and readable.
- Changing date range must update all dependent metrics and charts consistently.

8.13 Notifications and reminders

- Before requesting Android notification permission, show a clear rationale: what reminders do and how the user benefits.
- Trigger the platform permission request only when appropriate and only if the platform/version requires it.
- If permission is denied, keep the app usable and show a settings path to enable it later.
- Support daily goal reminders, study-task reminders, exam reminders, and admission milestone reminders where scheduling is available.
- Let the user toggle reminder categories and choose reasonable reminder timing.
- Store reminder preferences and scheduled-event identifiers so reminders can be updated or cancelled rather than duplicated.
- When an event is edited or deleted, update/cancel its pending reminder.
- Do not claim notifications are enabled or scheduled until the platform confirms success.
- Handle Android notification permission/version differences and exact-alarm restrictions where relevant to the existing implementation.
- Provide a Notification Center only for actual app-generated notification records; otherwise use a clear empty state.

8.14 Profile and settings

- Display saved avatar, preferred name, academic path, and accurate study summary.
- Allow editing name, avatar, academic details, and daily goal.

- Provide settings for appearance, notifications, reminders, privacy/data controls, and help/about as supported.
- If a theme setting already exists, it must work consistently across the whole app; this PRD's visual target remains all light mode.
- Never show fake streaks or totals. If there is no activity, show zero or a clear first-use prompt.
- Any reset/delete-data action must explain consequences and require confirmation.

9. Data model and persistence requirements

Adapt these entities to the existing codebase. Do not blindly replace an established schema; map the requirements to current models and document any migration.

Entity	Important fields	Relationships / rules
UserProfile	id, displayName, avatarId, academicLevel, group, targetYear, admissionTrack, dailyGoalMinutes, createdAt, updatedAt	One active profile per local account/session unless the app already supports authentication.
Subject	id, userId, name, banglaName, colorToken, iconKey, sortOrder, createdAt, updatedAt	Belongs to user; deleting requires clear linked-record policy.
Chapter	id, subjectId, title, sortOrder, status, notes, updatedAt	Status enum; progress calculated from chapter records.
StudyTask	id, userId, title, dueAt, durationMinutes, subjectId, chapterId, status, priority, notes, createdAt, updatedAt	Optional subject/chapter relation; status changes persist.
StudySession	id, userId, subjectId, chapterId, taskId, startedAt, endedAt, elapsedSeconds, status, sessionType	Actual elapsed time; protect against duplicate completion records.
AdmissionTarget	id, userId, institutionName, category, programUnit, cycleLabel, applicationStatus, notes, officialUrl, sourceType	Multiple targets per user; user can edit/archive.
AdmissionMilestone	id, targetId, type, title, dueAt, status, notes, reminderEnabled	Milestone types include open/deadline/admit card/exam/result/custom.
AdmissionChecklistItem	id, targetId, title, isComplete, sortOrder, completedAt	Progress percentage is derived from checklist items.
StudyNote	id, userId, subjectId, title, body, tags, isPinned, createdAt, updatedAt	Subject optional; search indexes as supported.
ReminderPreference	id, userId, category, enabled, leadTimeMinutes, preferredTime	Store permission state separately from user preference.
ScheduledReminder	id, relatedEntityType, relatedEntityId, platformNotificationId, scheduledAt, status	Update/cancel on entity changes; platform-specific.
AppSettings	userId, themePreference, locale, firstLaunchComplete, onboardingVersion	Persist safely and support migrations.

Persistence rules

- User-created data must survive app restart and normal app updates.
- Use the project's existing reliable local database/storage where available.
- If offline-first is promised, core create/edit/read flows must work without network access.
- If cloud sync is implemented, define conflict resolution and never silently overwrite newer data.
- Do not log private notes or sensitive profile data unnecessarily.
- Use schema migrations for data model changes; do not wipe user data during redesign.
- Keep timestamps consistent and define local-time display versus stored timestamp behavior.

10. Architecture and implementation constraints

- First inspect the actual project runtime and architecture. Do not assume it is Flutter, native Android, or a web app without checking.

- Use the existing framework and navigation structure unless a migration is necessary and justified.
- Create shared design tokens and reusable components rather than implementing each screen independently.
- Separate UI, domain/business logic, and persistence as appropriate to the current stack.
- Use one source of truth for profile, subjects, tasks, chapters, sessions, admission targets, theme, and reminder preferences.
- Use stable IDs for records and avatars.
- Use platform-native APIs for notification permissions and scheduling when the current runtime supports them.
- If the current environment cannot provide true native notifications, explain the limitation and do not simulate delivery as if it were real.
- Do not expose API keys, privileged secrets, or private service credentials in client code.
- Keep error states actionable and avoid silent failures.

Suggested implementation layers

Layer	Responsibilities
Presentation	Screens, reusable components, navigation, accessibility, responsive layouts.
Design system	Colors, typography, spacing, radii, icon mapping, motion, theme semantics.
Domain	Validation, progress calculations, timer state machine, admission countdown/status rules.
Persistence	Profiles, subjects, chapters, tasks, sessions, notes, targets, settings, reminders.
Platform integration	Notifications, app lifecycle, safe areas, keyboard, background timer behavior.
Testing	Unit tests for calculations, integration tests for workflows, UI review across screen sizes.

11. Business rules and calculations

Metric / rule	Calculation
Subject completion	Completed chapters / total chapters × 100; if total is zero, show not set up rather than 100%.
Overall syllabus progress	Completed chapters across included subjects / total chapters across included subjects; define whether archived subjects are excluded.
Today's study time	Sum actual elapsed seconds for eligible sessions in the current local calendar day; convert to hours/minutes for display.
Daily goal progress	Today's eligible study duration / daily goal duration × 100; cap ring at 100% but show actual duration.
Days remaining	Calendar-day difference between current local date and saved event date; handle past-due events distinctly.
Admission checklist progress	Completed checklist items / total checklist items × 100; if no items, show “Not started” or no progress, not 100%.
Streak	Consecutive local calendar days with activity meeting a documented threshold; no streak without qualifying saved activity.
Subject study time	Sum of eligible study-session elapsed seconds linked to that subject for selected range.
Upcoming event	Earliest non-cancelled future milestone with a valid date; exclude archived targets as defined.

Document one shared implementation for each calculation and use it across Home, Progress, Subject detail, and Profile. Avoid separate UI-only calculations that disagree.

12. Error, loading, and empty states

Scenario	Required response
No subjects	Show a clean illustration or icon, a concise explanation, and Add Subject action.
No tasks today	Show “Nothing planned yet” and Add Task / Plan Study actions.
No admission targets	Show a clear invitation to add a university and explain what can be tracked.
No sessions	Show zero study time and a Start Focus action; no fake chart bars.
No notes	Show Create Note action.
Save failure	Preserve user input, explain failure briefly, allow retry.
Notification permission denied	Explain reminders are off and provide a route to system settings where available.
Missing avatar asset	Use a bundled fallback illustration and keep profile stable.
Invalid/expired event date	Show an overdue/past state; never display negative “days left” without explanation.
Offline / unavailable service	Keep local functions usable where possible and explain unavailable cloud-only features.

Empty states should look designed, not like errors. Use the same illustration style, typography, spacing, and button hierarchy as the rest of the app.

13. Accessibility, localization, and Bangladesh context

- Use English as the initial UI language unless the existing product already supports a language setting; preserve the ability to support Bangla later.
- Ensure Bengali glyphs render correctly in profile fields, subject Bangla names, and notes.
- Use local date/time presentation while storing dates/times in a consistent format.
- Do not assume all users have high-end devices or always-on internet.
- Support Android font scaling, meaningful content descriptions, sufficient contrast, and adequate touch targets.
- Avoid color-only status indicators; combine color with labels or icons.
- Make forms keyboard-safe and allow scrolling to focused inputs.
- Use clear, student-friendly language without shame-based or manipulative progress messaging.

14. Security, privacy, and data integrity

- Collect only information needed for the study-management experience.
- Do not request unrelated permissions.
- Do not transmit profile, notes, or academic records to an external service unless the feature requires it and the user is informed.
- Use secure storage for secrets/tokens if any exist; do not hardcode privileged keys.
- Confirm destructive actions and explain their impact.
- Avoid unnecessary analytics containing personal notes or sensitive user-entered text.
- Provide data export/reset behavior only if it can be implemented reliably; clearly describe what is included.
- Do not claim data is synced or backed up unless a real sync/backup mechanism exists.

15. Functional acceptance criteria

ID	Area	Pass condition
AC-01	First launch	Fresh install shows splash then onboarding; does not jump directly into a hardcoded demo profile.

AC-02	Returning launch	After successful setup, relaunch opens the app without repeating Get Started/onboarding.
AC-03	Avatar persistence	Selected cartoon avatar appears in Home and Profile after force-close/reopen.
AC-04	Academic branching	SSC/HSC/admission selections show only relevant fields and save correctly.
AC-05	Subject creation	Create Subject saves the subject; it appears in Syllabus and Home; preview matches selected name/color/icon.
AC-06	Chapter progress	Changing chapter status updates subject and overall progress using real counts.
AC-07	Task persistence	Create/edit/complete a task; data remains correct after restart.
AC-08	Focus timer	Start/pause/resume/finish follows a valid state machine and creates at most one session record.
AC-09	Study time	Home and analytics show the same eligible total for the same date range.
AC-10	Admission tracking	Add two targets with different milestones; calendar and upcoming list display both correctly.
AC-11	Countdown	Days remaining derives from saved milestone date and updates with local date.
AC-12	Notification permission	Rationale precedes request; denial leaves app usable and reminders are not falsely reported as active.
AC-13	Reminder updates	Editing/deleting an event updates or cancels its scheduled reminder when supported.
AC-14	Light visual consistency	Every redesigned screen uses the shared light-mode palette and components.
AC-15	Topbar	Main header has premium StudyHub branding and notification access, with no hamburger.
AC-16	Safe areas	Content is not obscured by status bar, keyboard, bottom navigation, or gesture area.
AC-17	Empty states	No fake metrics or fake records appear for a new user.
AC-18	Navigation	Every visible primary action leads to a working screen or a clearly documented unavailable capability.
AC-19	Data preservation	Existing user records are retained or migrated safely; redesign does not silently reset data.
AC-20	Build quality	Build/test command completes successfully where the environment permits; any blocked tests are reported honestly.

16. UI quality acceptance criteria

- The latest embedded all-light mockup is used as the primary visual target.
- All screens share the same app background, surface styling, typography scale, border treatment, icon family, and spacing rhythm.
- No dark screen is mixed into the light UI showcase or redesigned flow.
- No hamburger icon appears in the main app header.
- The StudyHub wordmark is bold, recognizable, and consistent on splash and main header.
- Cartoon avatars are cohesive original/bundled assets and not letter-only circles.

- Subject creation sheet matches the premium form quality shown in the reference: preview, fields, color swatches, icon selection, and strong CTA.
- Admission calendar and progress screens feel as polished as Home, not like unfinished admin forms.
- Bottom navigation has consistent active/inactive states and stays at the bottom above system insets.
- Cards and buttons do not jump or resize unexpectedly during common interactions.
- The UI remains usable with long names, long institution titles, no data, and larger text settings.

17. QA test plan

Test group	Test cases
First launch	Fresh data; onboarding completion; back navigation; app kill during setup; resume state.
Profile	Edit name, avatar, academic level, goal; restart app; verify persistence.
Subjects	Create, edit, duplicate warning, delete confirmation, long names, Bangla name, icon/color preview.
Chapters	Add, reorder if supported, mark each status, verify progress math and empty behavior.
Planner/calendar	Create tasks around month boundaries, edit date, complete task, filter statuses, open detail.
Admission	Multiple targets, missing optional dates, overdue date, deadline reminder, status changes, archive/delete.
Focus timer	Start, pause, resume, app background, finish, cancel, repeated taps, interruption.
Analytics	7-day/30-day/3-month ranges, no activity, activity across midnight, subject breakdown consistency.
Notifications	Permission granted, denied, revoked in settings, schedule/update/cancel, Android version behavior.
UI layout	Small phone, large phone, keyboard open, long text, increased font size, safe areas, scrolling.
Visual consistency	Compare each screen to the embedded reference; verify no dark surfaces, random colors, or inconsistent icon styles.
Regression	Existing working routes, persistence, and features remain functional after redesign.

18. Implementation sequence

The product may be implemented as a broad end-to-end delivery, but it must be built and verified in coherent slices. Do not produce a huge untested patch and declare success.

Phase	Scope	Exit gate
P0 — Audit	Inspect project, framework, navigation, current data, theme, notifications, broken flows.	Written inventory of current architecture and risks; backup/version checkpoint.
P1 — Design foundation	Tokens, typography, logo/wordmark, shared header, bottom navigation, cards, fields, buttons, sheet, avatar component.	Core components render consistently in the existing project.
P2 — First-run experience	Splash, onboarding, academic branching, avatar persistence, goals, routing.	First launch and returning launch both pass.
P3 — Study foundation	Subject CRUD, chapters, syllabus, Home calculations, empty states.	Subject/chapter state persists and progress is correct.
P4 — Planner and admission	Tasks, calendar, admission targets, milestones, deadlines, status, countdown.	Multiple events and targets render correctly from saved data.
P5 — Focus and insights	Timer, sessions, notes, analytics, profile metrics.	Study time and charts derive from saved records.
P6 — Notifications	Permission rationale, permission handling, schedules, update/cancel, settings.	Actual supported platform behavior is tested; limitations documented.

P7 — Polish and QA	Motion, accessibility, safe areas, responsiveness, regression and visual review.	Acceptance criteria pass; remaining issues listed honestly.

19. Instructions for the AI coding agent

Use the following implementation contract when this PRD is attached to the existing project.

```
You are the senior engineer responsible for implementing StudyHub from the attached Master PRD and
embedded all-light UI reference.

SOURCE OF TRUTH
1. This PRD defines product behavior, functional requirements, calculations, data rules, and
acceptance criteria.
2. The embedded StudyHub mockup defines the visual direction and screen composition.
3. Existing working code and user data must be preserved unless a documented change is necessary.

MANDATORY VISUAL CONSTRAINTS
- Rebuild the reference as real responsive screens; never use the collage/screenshot as a screen
background.
- ALL redesigned screens must follow the same LIGHT MODE design system: pale near-white
background, white surfaces, navy text, restrained teal/blue accents, consistent rounded cards,
subtle borders, refined typography, and cohesive icons.
- Do not insert any dark-mode screen, dark dashboard, black card, or mixed light/dark UI into the
redesigned flow.
- Remove the hamburger from the primary header. Build a premium StudyHub wordmark and logo, with
notification access on the right.
- Keep the bottom navigation at the bottom of app content, above Android system insets.
- Use a cohesive set of bundled cartoon avatars, not letter-based placeholders or random external
images.
- Use shared tokens and reusable components across all screens.

FUNCTIONAL CONSTRAINTS
- Inspect the actual project and identify its framework before editing.
- Implement real data flows and persistence; no fake metrics, hardcoded demo profile, invented
admission dates, fake chart bars, or dead buttons.
- Implement first-launch onboarding and persist completion; returning users must not see Get
Started repeatedly.
- Ensure subject/chapter/task/focus/admission/calendar/notes/progress flows use real state.
- Request notification permission only after explaining why; handle denial and platform
limitations honestly.
- Preserve existing working features and data. Do not migrate frameworks unless justified.
- Do not claim builds, tests, or platform features passed unless actually verified.

WORKING METHOD
1. Audit project and report current framework, entry points, navigation, persistence, notification
implementation, and likely risks.
2. Implement shared design system and app shell.
3. Implement onboarding and profile persistence.
4. Implement study features and admission tracking.
5. Implement reminders and analytics.
6. Run build/tests, fix regressions, and compare all screens against the reference.
7. At completion, report changed files, completed features, actual tests run, known limitations,
and any remaining work.

Do not stop after describing a plan. Make changes in the existing project. If a platform or tool
limitation prevents a requirement, explain it and implement the best real alternative rather than
faking it.
```

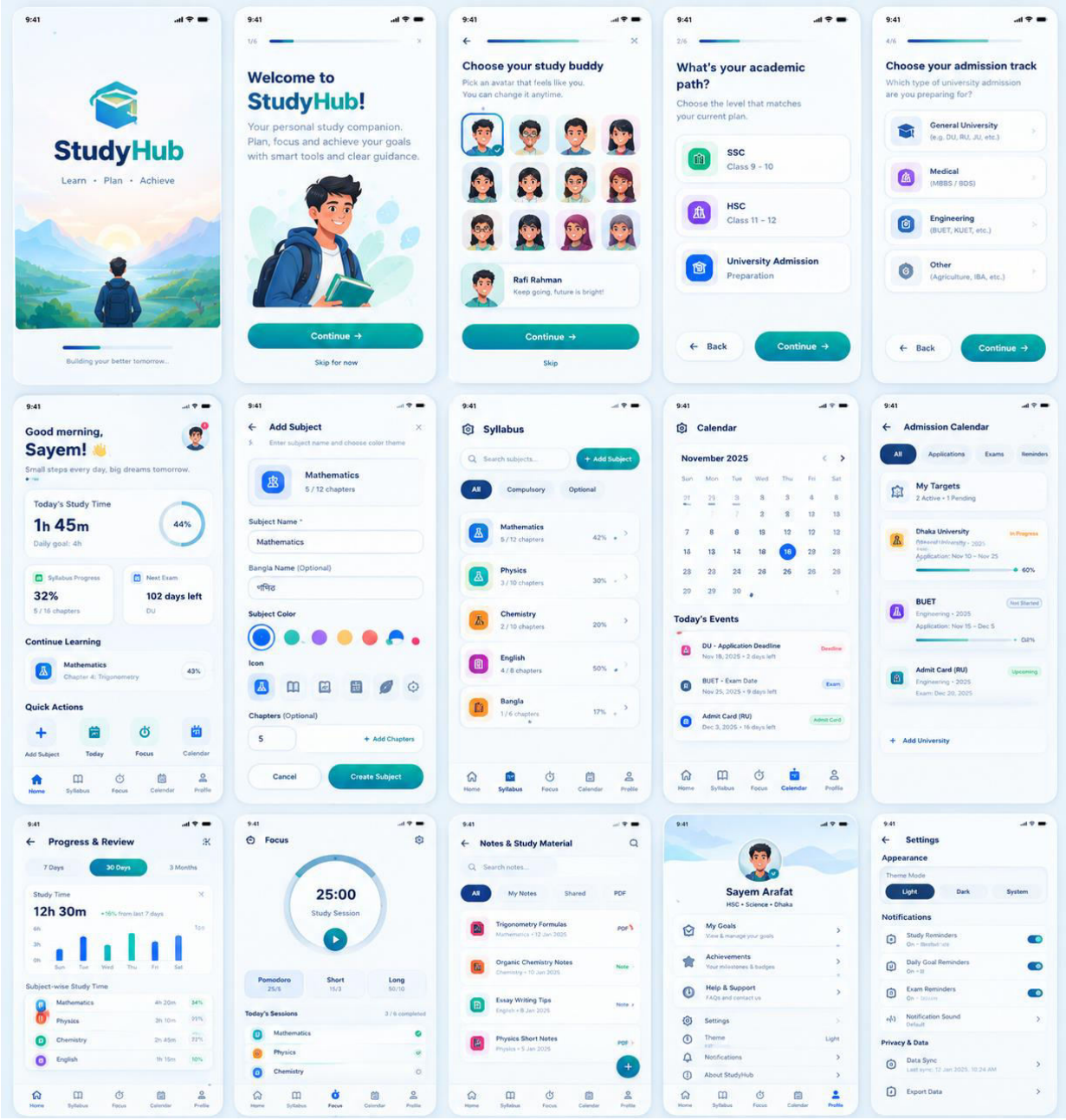
20. Release checklist

- Brand mark, StudyHub wordmark, splash, header, and avatar assets are consistent.

- All screens in the redesigned flow use the light visual system.
- No hamburger remains in the primary topbar.
- First-launch onboarding works and does not repeat on returning launch.
- Profile, academic path, goal, and avatar persist.
- Subject and chapter CRUD works and progress calculations are correct.
- Planner and calendar show actual saved tasks and dates.
- Admission targets support multiple institutions and multiple milestones.
- Focus sessions record real elapsed time with duplicate protection.
- Analytics match the underlying session and chapter records.
- Notification permission rationale, grant, denial, and settings recovery are tested.
- No fake demo metrics, hardcoded countdowns, or fictional admission dates remain in production paths.
- All key controls are reachable and accessible.
- System bars, keyboard, safe areas, and bottom navigation are correct.
- Build/tests and limitations are reported truthfully.

Appendix A — Reference UI image

This image is the primary visual reference for the StudyHub redesign. Keep the design language consistent across every screen and do not mix in dark screens.



Appendix B — Key product copy

Context	Suggested copy
Welcome	Your personal study companion. Plan, focus and achieve your goals.
Avatar selection	Choose your study buddy.
Academic level	What's your academic path?
Admission track	Choose your admission preparation track.
Daily goal	How much time do you want to study each day?
Empty subjects	Build your study plan one subject at a time.
Empty admission calendar	Track application deadlines, admit cards, and exam dates in one place.
Empty study history	Start a focus session to see your progress here.
Notification rationale	Enable reminders so important study tasks and exam dates don't slip by.

Appendix C — Decisions that must not be guessed

- Do not guess the current framework, database, backend, or notification capabilities; inspect the project.
- Do not guess official university admission dates. Use user-entered dates or verified sources with explicit provenance.
- Do not assume that permission granted means a reminder was successfully scheduled.
- Do not assume a visual percentage is correct without a defined calculation.
- Do not assume the user wants dark mode in the mockup; this reference and redesign target are all light mode.
- Do not replace existing user data or working features to simplify the redesign.